



Architectural **Portfolio**

Yonatan Demssie
Selected Works 2024 – 2025



About Me

I am an architectural designer focused on computational design and research-driven methodologies. My work investigates the intersection of performance and spatial experience, specifically within complex typologies including healthcare infrastructure, vertical density, and urban renewal.

I thrive on the complexity of large-scale projects because they offer an immense learning curve, requiring a comprehensive understanding of how diverse systems integrate. I view architecture fundamentally as a service a discipline requiring technical rigor, environmental stewardship, and empathy

Yonatan Demssie

- ✉ Yonathan.Yoseph@icloud.com
- ☎ +17202990461
- 🌐 www.linkedin.com/in/yonatan-demssie
- 📷 https://www.instagram.com/yanatan_yoseph/
- 📺 www.tiktok.com/@53_design_studio

Education

- JUNE 2021 | Unity University — Addis Ababa, Ethiopia
(B.Sc.) Architecture and Urban Planning & Design
- JAN 2026 | Cumulative GPA: 3.75 | Major GPA: 3.96
Graduation Thesis: Pediatric Oncology Hospital

Professional Experience

- JUNE 2024 | Convention center (Design Proposal)
Parametric High-rise (Design)
AA Tower (Design Proposal)
Convention center (Design Proposal)
Water Park
Hagerly Assets (non Profit)
Contributed to major design decisions across ongoing architectural projects under direct mentorship from firm principals.
Developed advanced 3D models, parametric studies, and presentation materials for real-world projects.
Applied computational design workflows to support form exploration, and design lderation.
- OCT 2024 | Ahaa Consulting Architects & Engineering

PROJECT EXPERIENCE

- JUNE 2025 | Pediatric oncology hospital 8
- Hyundai Regional HQ 20
EV Battery manufacturing
Urban Planning and Design V
Advanced Landscape Architecture
Interior Design 26
- internship at Ahaa 32
Fine Arts School
Landscape Architecture
Low-Cost Building Strategies
Urban Planning and Design iii
Mass Housing
Architectural Science (Acoustics)
Urban Planning and Design ii
Apartment Complex
Urban Planning and Design i
Theory of Architecture
Architectural Science (Heating Cooling And Vent)
Residence
Architectural Science (Light)
History Of Ethiopian Architecture
- OCT 2022

Softwares & Skills

DRAFTING & BIM

Revit 
AutoCAD 

3D MODELING

Rhino 3D 
Sketchup 
3ds MAX 

COMPUTATIONAL

Grasshopper 
Rhino Inside 

RENDERING

D5 Render 
Enscape 
Lumion 
Chaos v-ray 
Chaos Corona 

PRESENTATION

InDesign 
Illustrator 
Photoshop 
Premiere Pro 
DaVinci Resolve 

Advanced

Beginner

01 Graduation Project



Pediatric Oncology
Hospital

0 06

02 High Rise



Hyundai Regional
HQ

0 20

03 Interior Design



Japanese
Bar & Restaurant

0 26

04 Internship



Convention Center
Water park

0 32



Adiss Abeba
Tower

0 36



Adiss Abeba
Convention Center

0 38

Academic

Selected works

Professional

St. Paul's Pediatric Oncology Hospital

Light to heal

St. Paul's Pediatric Oncology Hospital is a specialized healthcare facility proposed as an extension of St. Paul's Hospital Millennium Medical College in Addis Ababa, Ethiopia. The project establishes a dedicated environment for long-term pediatric oncology care while remaining institutionally connected to the existing medical campus

Developed as a **Bachelor of Architecture graduation project**, the proposal investigates how architectural design can respond to the complex clinical, emotional, and operational demands of pediatric oncology within a resource-constrained healthcare context.



Area	Designer	Type	Location
4.2Ha 10.4ac	Yonatan Demssie	Hospital	Addis Ababa Ethiopia

Documentation

Design

Presentation



Areal Perspective

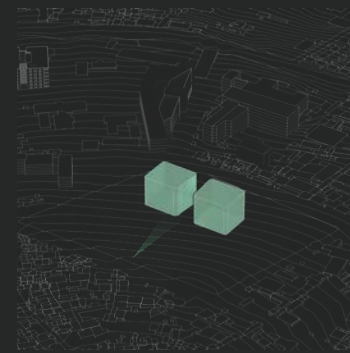
Light to heal



St. Paul Millennium
Medical College

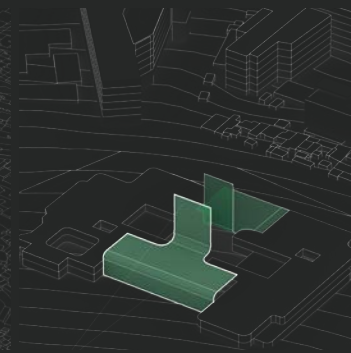
Oncology
Department

St. Paul's
Hospital



Cut for

View light
Clarity of orientation



Atrium for

Volume
space within a space
openness and legibility



Circulation

Looping
Hide service Route
Hide The mansion part



Public realm

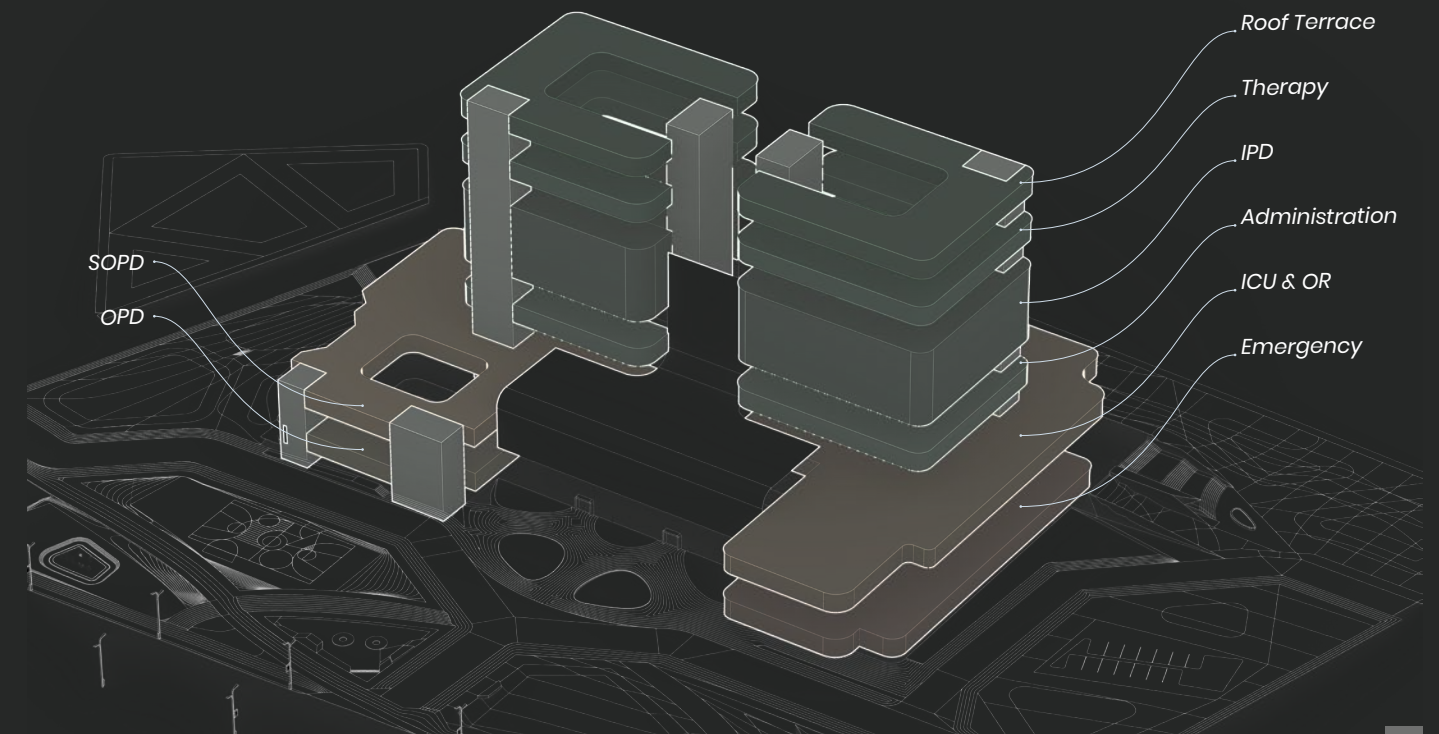
Journey
Fenseless Architecture
Experience Design For All
Stakeholders

Core Drivers

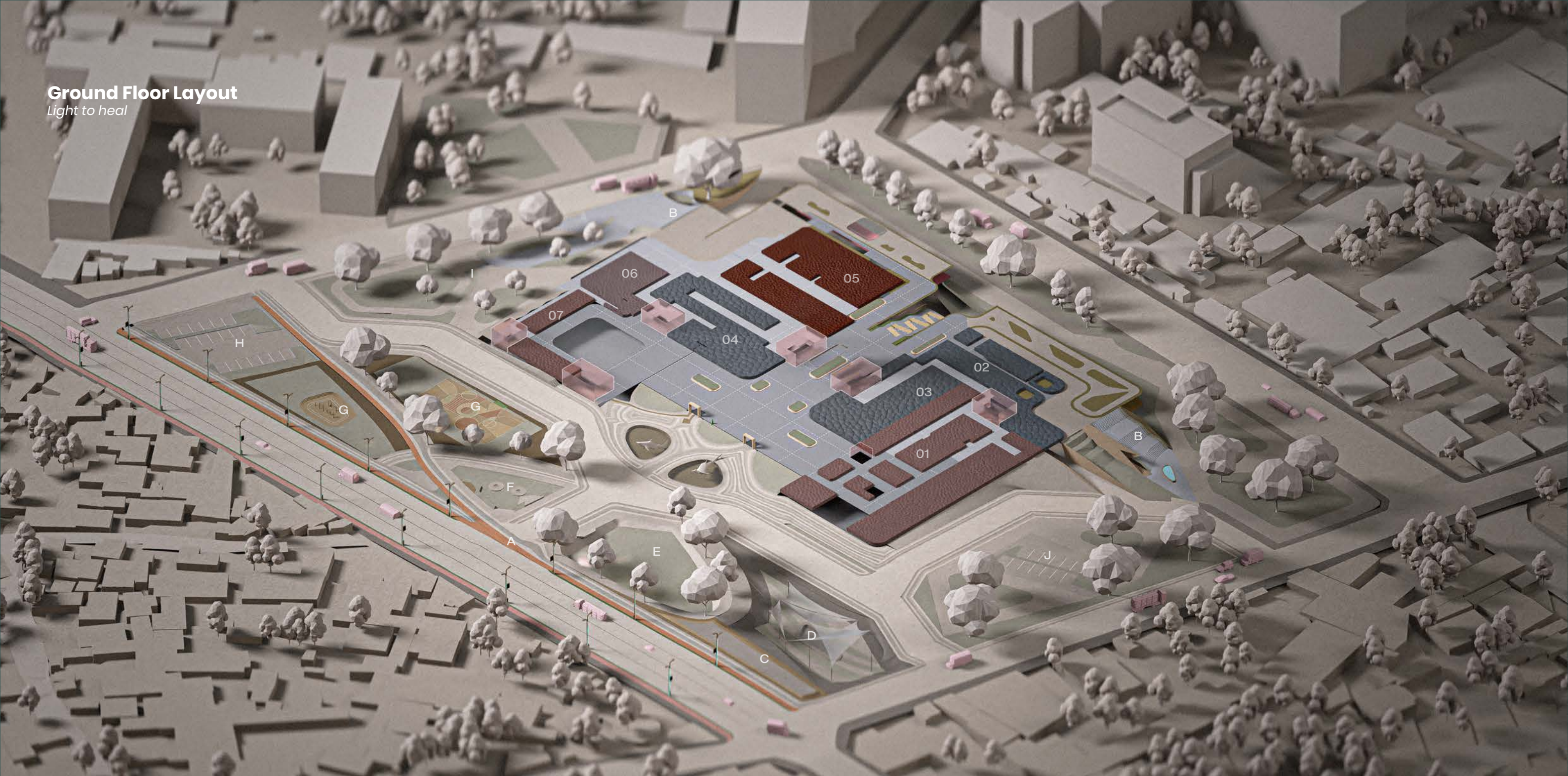
Light to heal

Program Diagram

Light to heal



Ground Floor Layout
Light to heal



A Main Entrance

C temporary Shops

E Event lounge

G Play ground

G Park

01 Emergency

03 Pharmacy

05 Kitchen & Restaurant 07 OPD

B Staff Entrance

D Green House

F flag post

H OPD Parking

H Emergency parking

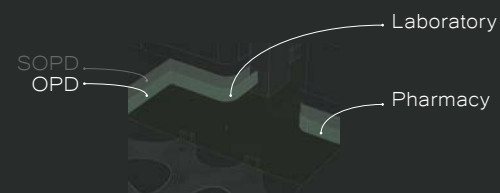
02 Admission

04 Laboratory

06 Daily procedure

08 Daily procedure

Isometric
View NW



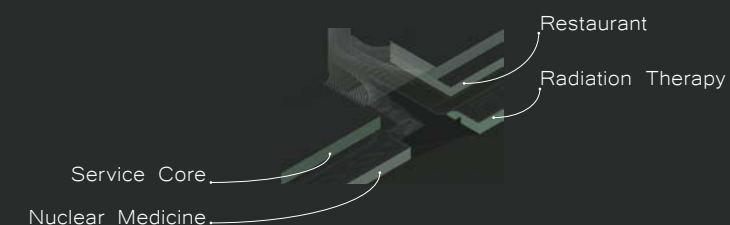
Space with in a space
Atrium solution to

Central Atrium: Provides visual clarity, orientation, and direct connection across clinical zones.

Safe Play Zone: Enclosed, supervised space for pediatric mobility and social play.

Healing Environment: Water features & greenery soften the clinical atmosphere and support emotional well-being.

Isometric
View SW



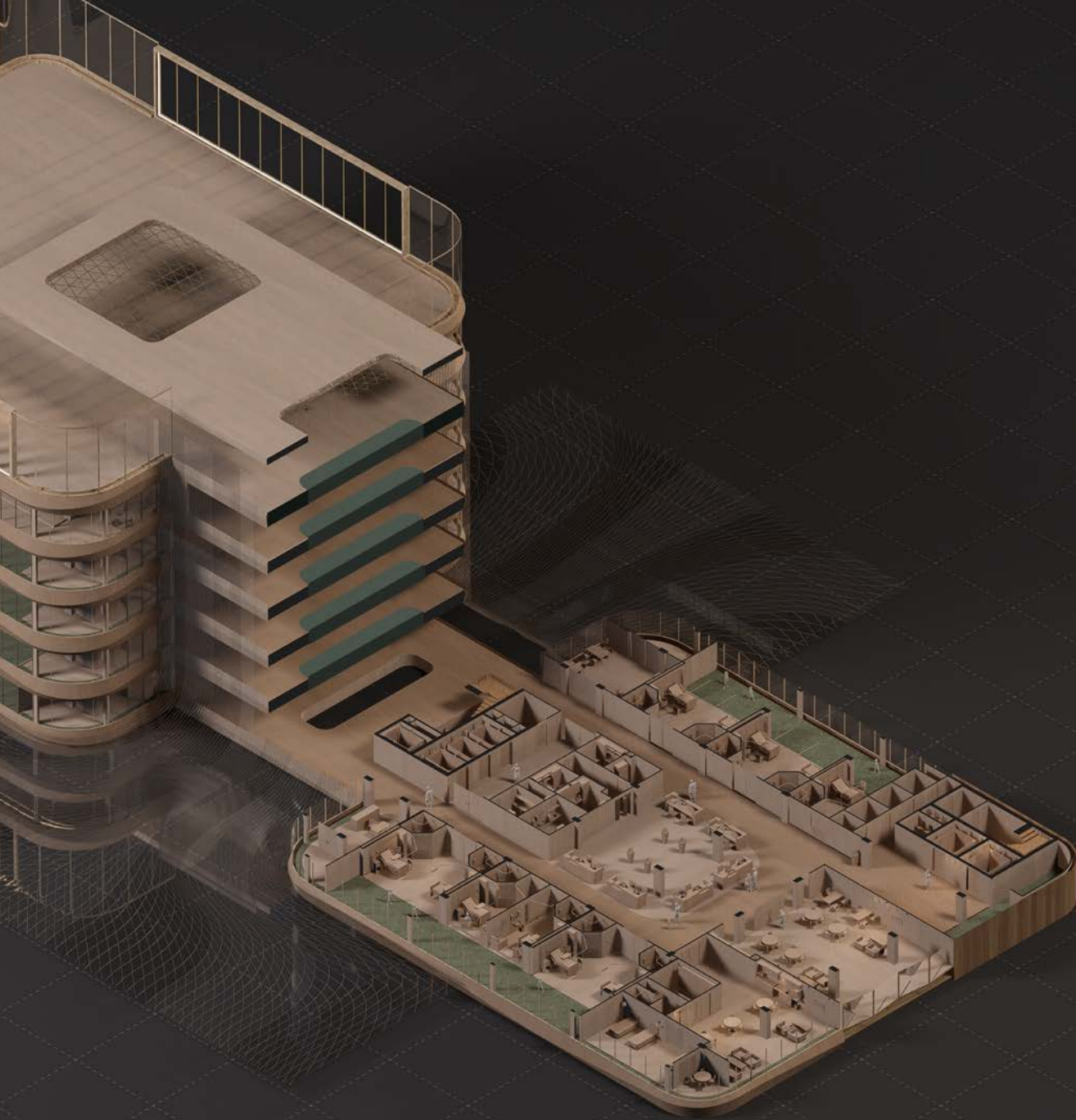
Stair-ramp solution to

Service Continuity: Vertical core consolidates sterile, staff-only, and technical pathways.

Stair-Ramp Sequence: Creates an intuitive, accessible patient journey with controlled transitions.

Functional Integration: Radiation, imaging, and nuclear medicine positioned for safe, efficient circulation.

Perspective Section Diagram
Light to heal



IPD Floor Layout

Light to heal

Double skin Facade solution to

- **Passive Solar Regulation:** Deep horizontal shading devices mitigate solar heat gain while preserving external views.
- **Diffused Daylighting:** Facade geometry is optimized to increase the ambiance of natural light without causing glare.
- **Bibliophile Interface:** Integrated planter boxes provide patients with immediate visual and physical access to nature.
- **Social Cohesion:** Strategically placed communal spaces foster interaction to build camaraderie and support networks.

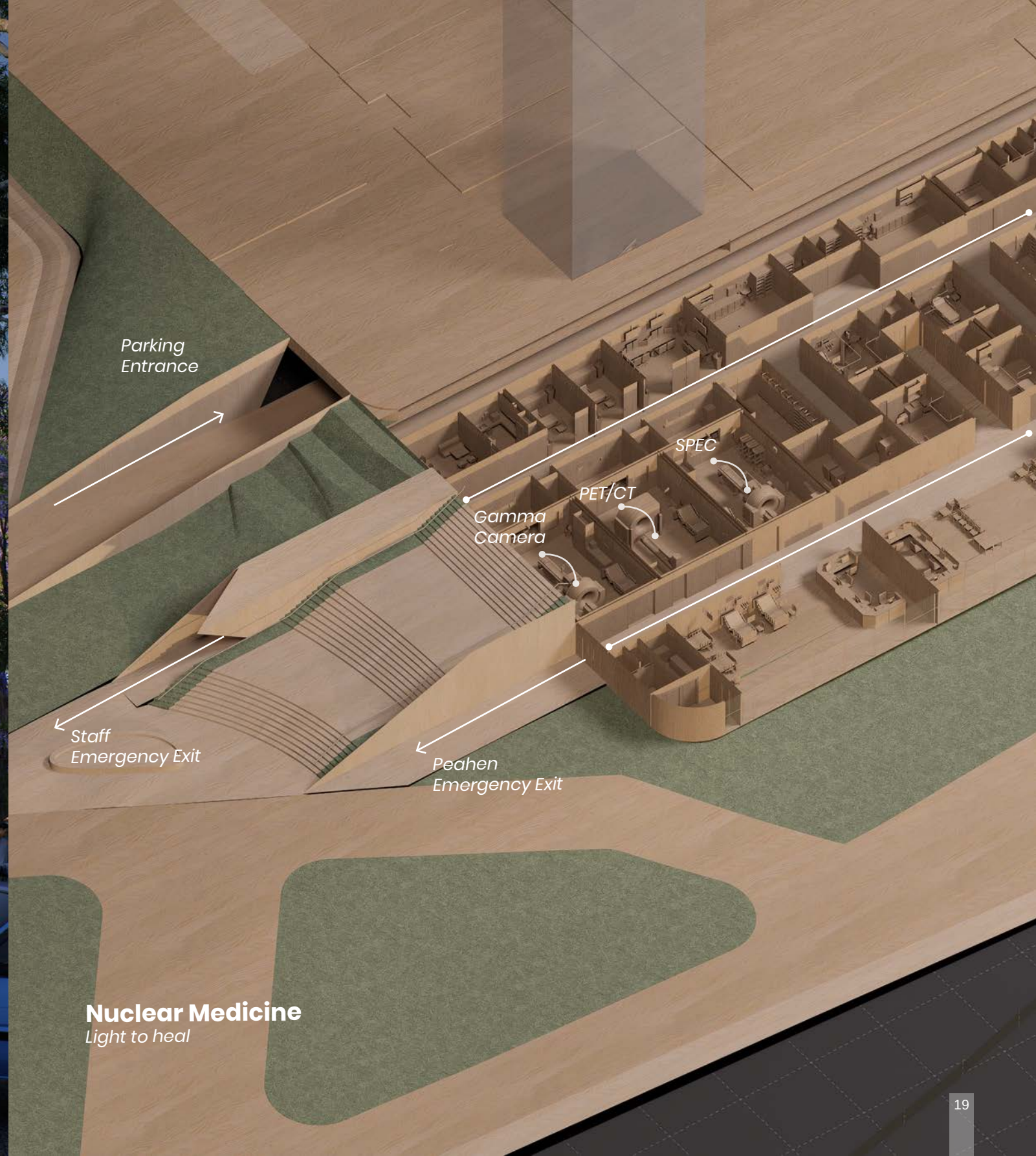


Double Skin Space Perspective

Light to heal



Staff Entrance Perspective
Light to heal



Nuclear Medicine
Light to heal

02

Hyundai Regional HQ High rise design

This project proposes a regional headquarters for Hyundai Motors and Marathon Motors, developed alongside an off-site EV battery manufacturing and recycling facility within a studio prescribing high-rise and industrial typologies. Conceived as a unified ecosystem, the design is informed by Meskel Daisy (the flower) a symbol of renewal across Ethiopia's 85+ ethnic groups bridging South Korean corporate precision with Ethiopia's cultural context through spatial, structural, and formal expression.



Area	Designer	Type	Location
1.2Ha 3ac	Yonatan Demssie	High-rise	Addis Ababa Ethiopia

Documentation

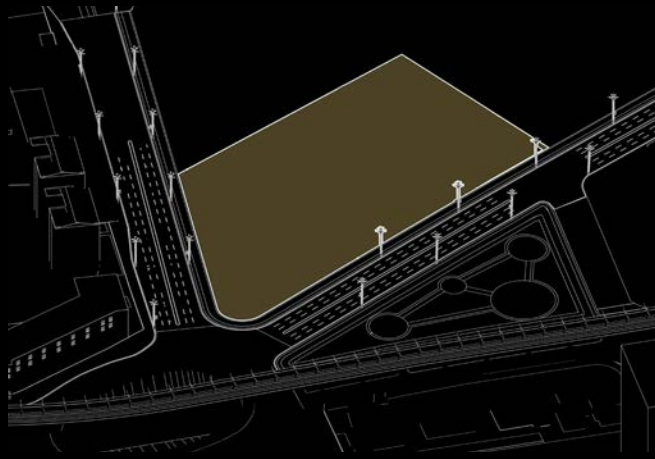


Design



Presentation

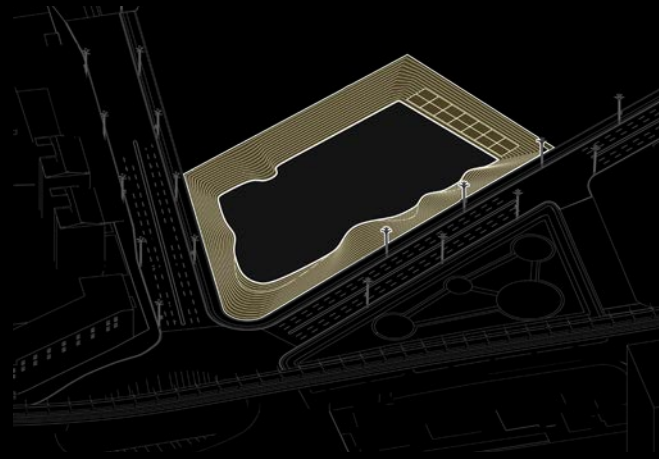




Site information

View
light

Clarity of orientation



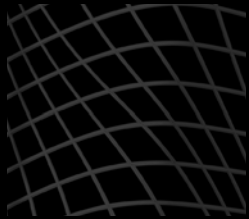
Urban Response

View
light

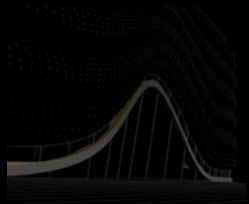
Clarity of orientation

Physical information

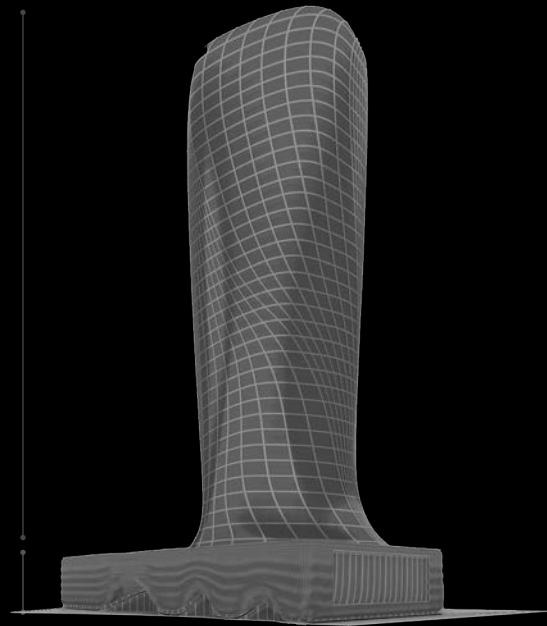
Metaphorical information



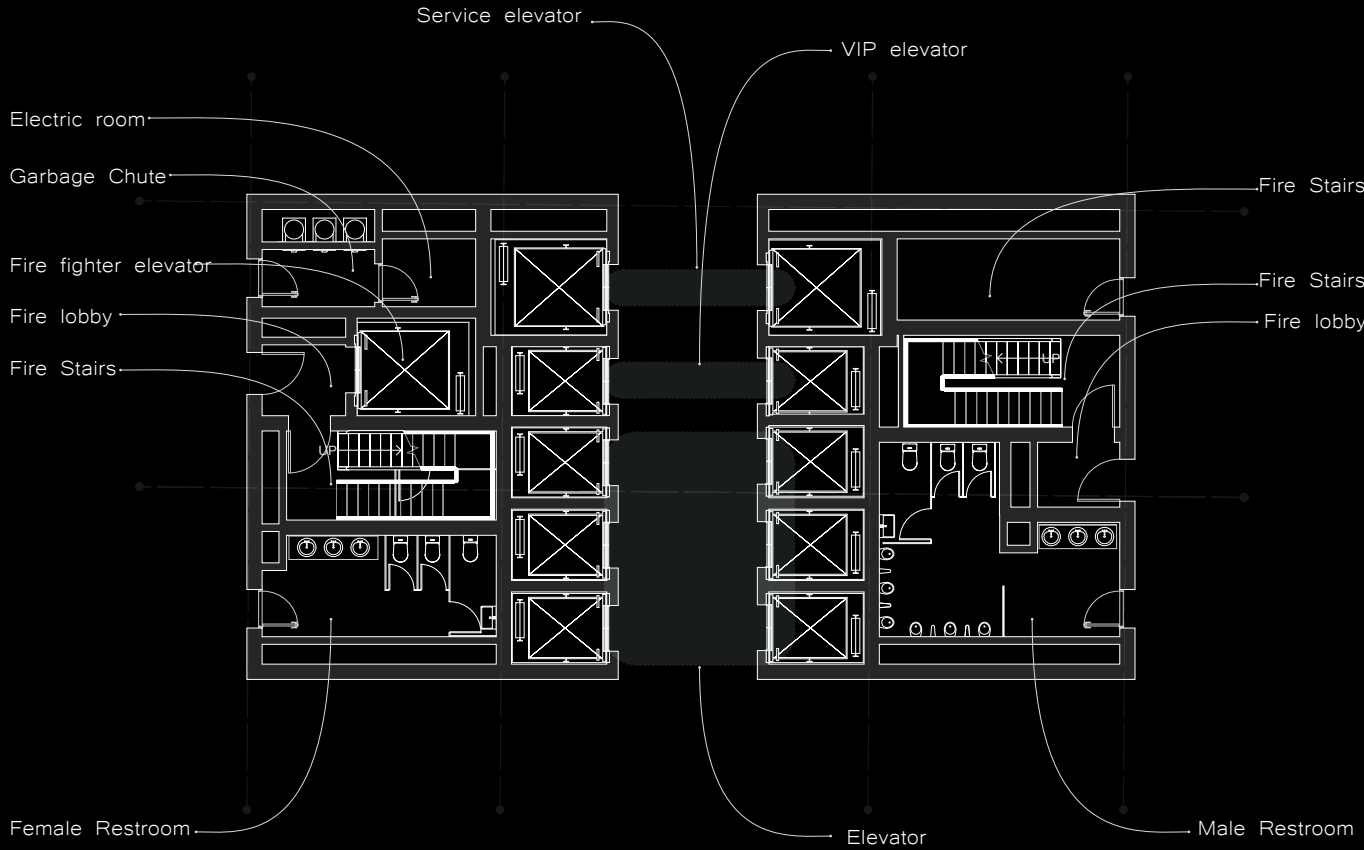
Expression



Exploration



Core Design
Meskel Dasie Flower



-
- The Program Diagram shows a vertical section of the Meskel Dasie Flower building, illustrating the distribution of various functional spaces across its height. The building is depicted with a curved, ribbed facade. The spaces are listed vertically on the left side of the diagram, with horizontal lines indicating their vertical extent within the building's structure.
- Restaurant
 - Penthouse
 - Staff housing
 - Hyundai offices
 - Staff services
 - Marathon motors Offices
 - R&D
 - Public Programs
 - Parking&services

Program Diagram
Meskel Dasie Flower

03

Interior Design Japanese Bar & Restaurant

This project was developed as the final design for an Interior Design studio that prescribed a restaurant typology. A Japanese bar and restaurant was selected based on contextual and stylistic studies appropriate to the site. The concept, "Let It Breathe," emerged in response to extreme spatial compression, where most walls rest directly on property boundaries and openings are introduced only to meet minimum daylight requirements mandated by code. The project critically addresses a front facade driven primarily by visual expression rather than spatial performance, reclaiming interior quality through restraint, light, and purposeful space-making.

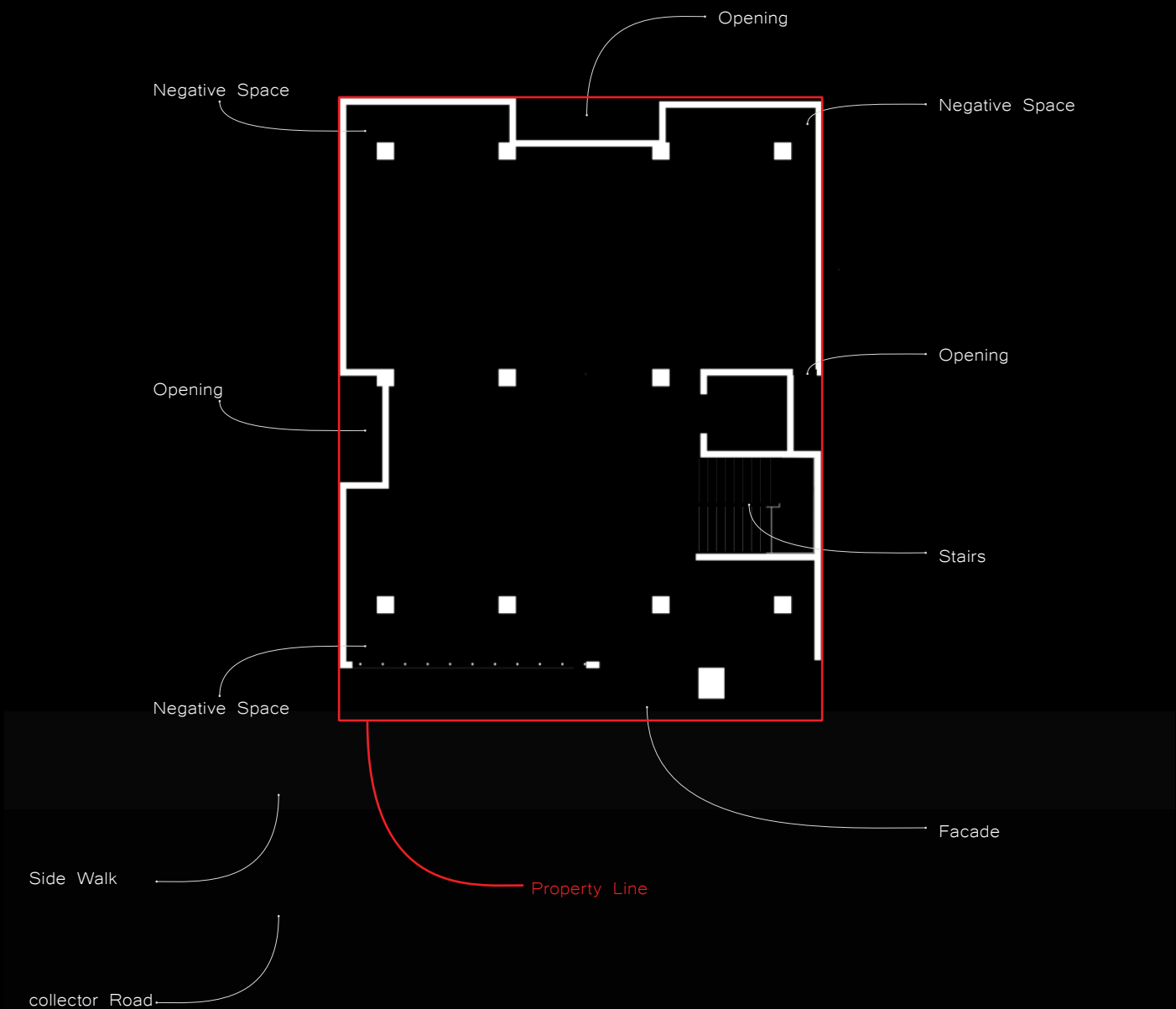


Area	Designer	Type	Location
312m2 3360sq.f	Yonatan Demssie	Interior	Addis Ababa Ethiopia

Documentation	Design	Presentation
		
		

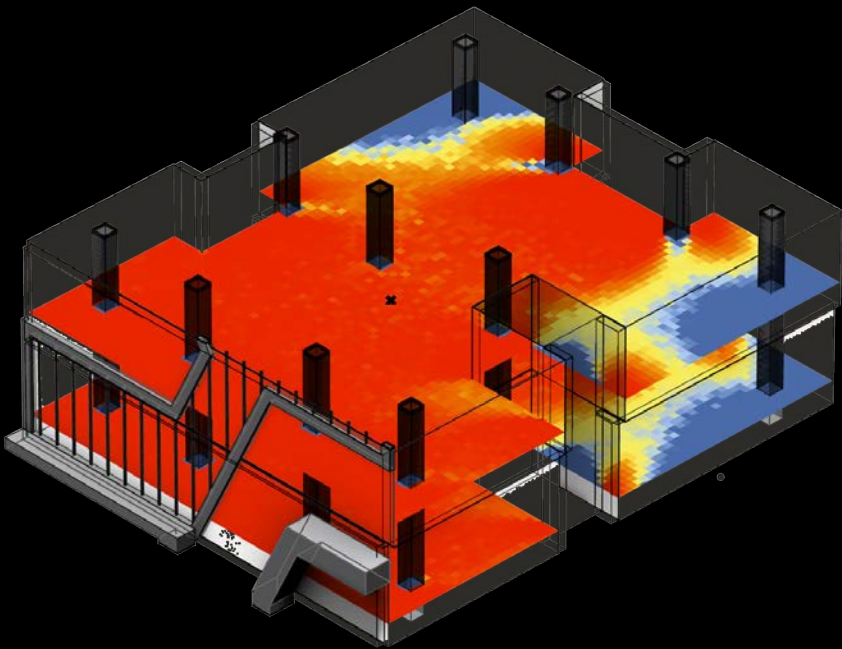


Site Conditions
Lt is bright



Sunlight Analysis
Lt is Bright

With Lady bug Tools



Sunlight analysis with Ladybug enabled iterative testing of design variations, quantitatively measuring interior daylight quality and guiding decisions based on environmental performance rather than purely aesthetic intent.



Major Design Destitution
Lt is bright



03 Internship at Ahaa Architects

Designers

Yonatan Demssie

Betsegaw A. Tamene

Managing partner at Ahaa

Documentation

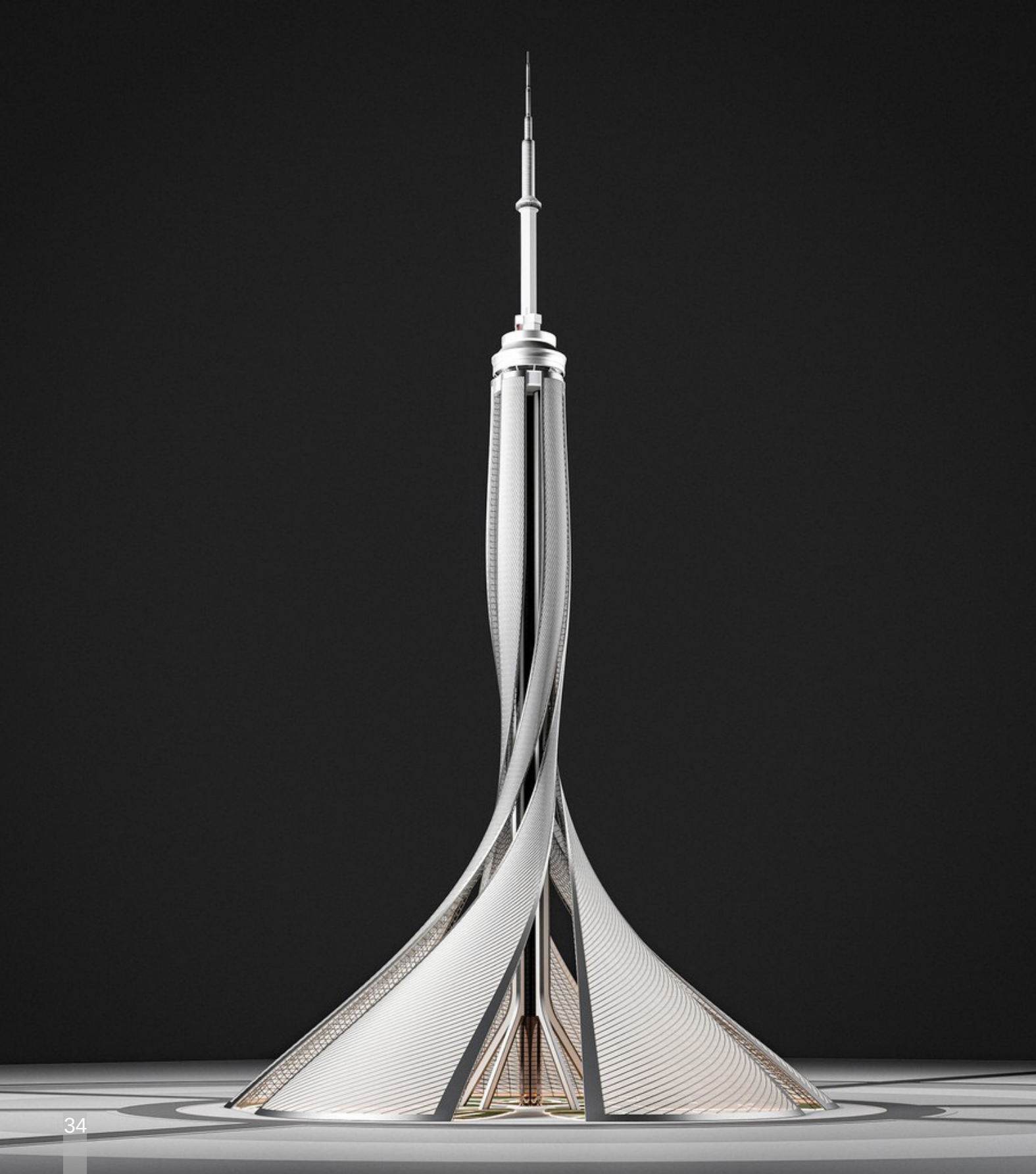


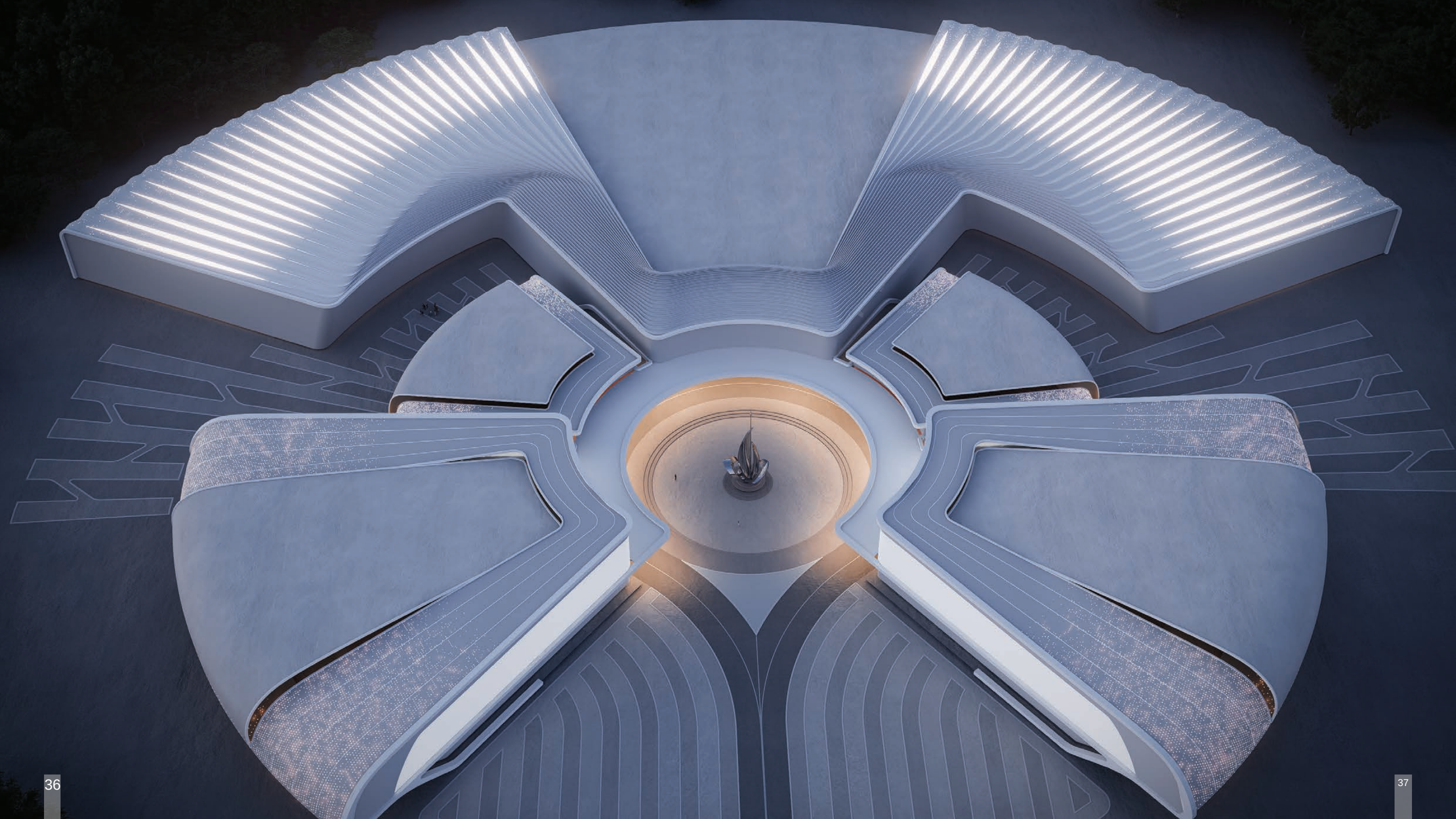
Design

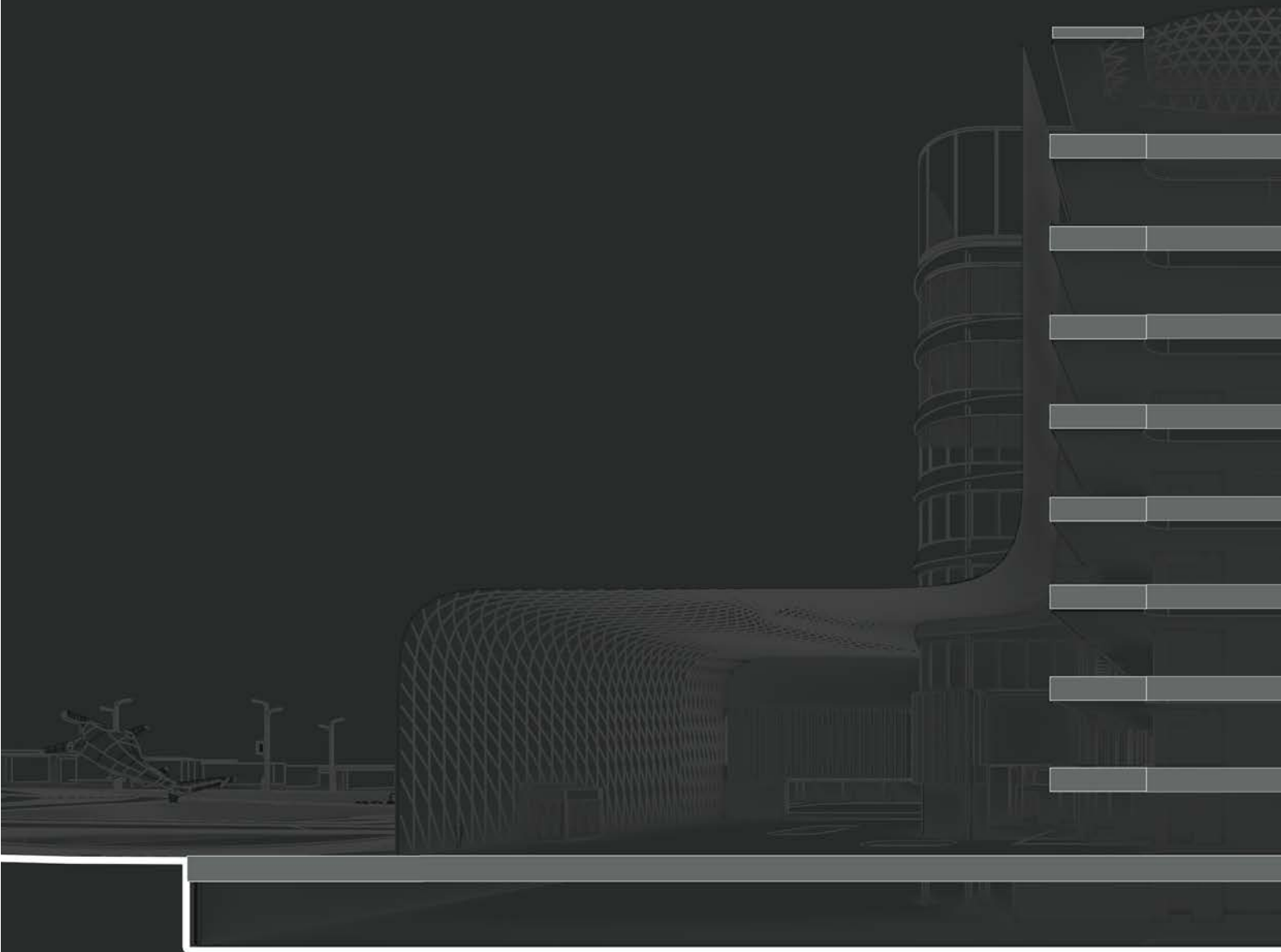


During my internship at AHA Architects, I worked within a highly supportive mentor ship environment that allowed me to meaningfully contribute to design development while growing under real-world professional conditions. I was exposed to major design decision-making processes and developed my skills across digital modeling, fabrication logic, and computational workflows. The studio culture encouraged research, experimentation, and the integration of design and technology, which significantly accelerated my technical confidence and design thinking. This period marked a clear shift in how I approach digital tools and architectural problem-solving. The work presented here highlights selected outcomes from this experience; **project content is limited to material that has been publicly disclosed.**









Thank You